Route-chart lifts over finite rings: $N(4n, 5) \geq 144\, n^5$ for every $n \not\equiv 2 \pmod 4$

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Abstract

The route-chart lift of arXiv:2606.15860 combines a 4-regular *controller* on 144 vertices with a *universal route chart* on $\mathbb{F}_q^5$ and gives graphs of maximum degree $4q$ and diameter at most 5 on $144\, q^5$ vertices. The paper gives universal charts for $q = 3$ and $q = 4$. We show that the construction works over any finite commutative ring that carries a universal chart, and that such rings exist in every order $n \not\equiv 2 \pmod 4$. Hence $N(4n, 5) \geq 144\, n^5$ for every $n \geq 1$ with $n \not\equiv 2 \pmod 4$, in particular for every prime power $q \geq 3$. We also show that universal charts exist over every finite field $\mathbb{F}_q$ with $q \geq 3$, and that none exists over $\mathbb{F}_2$. The first statement is formalized in Lean 4 with Mathlib, including all finite checks, which the Lean kernel verifies.

1 Results

Let $N(\Delta, D)$ be the largest order of a graph with maximum degree at most Δ and diameter at most D.

Theorem 1. *For every integer $n \geq 1$ with $n \not\equiv 2 \pmod 4$ there is a graph with $144\, n^5$ vertices and maximum degree at most $4n$ in which every two vertices are joined by a walk of length at most 5. Hence*

$$N(4n, 5) \;\geq\; 144\, n^5 \qquad (n \geq 1,\ n \not\equiv 2 \pmod 4).$$

Every prime power $q \geq 3$ is odd or divisible by 4, so:

Corollary 2. *$N(4q, 5) \geq 144\, q^5$ for every prime power $q \geq 3$.*

For $n \geq 2$ the order $144\, n^5$ exceeds the Moore bound $\Delta^4 - 2\Delta^3 + 2\Delta^2 + 1 < \Delta^4 = 256\, n^4$ for maximum degree $\Delta = 4n$ and diameter 4, so these graphs have diameter exactly 5. For $n = 3$ and $n = 4$ they are the graphs of arXiv:2606.15860 on 34,992 and 147,456 vertices. For $n = 5$ the graph has degree 20 and 450,000 vertices. Its edge list was checked directly, by breadth-first search from every vertex (Section 8).

The construction in arXiv:2606.15860 is stated for finite fields. Our second result says that the field version also works for every $q \geq 3$.

Theorem 3. *For every prime power $q \geq 3$ there is a universal route chart on $\mathbb{F}_q^5$ (Definition 7). With the 144-vertex controller, the route-chart lift over $\mathbb{F}_q$ is a $4q$-regular graph of order $144\, q^5$ and diameter 5.*

Remark 4 (How Theorems 1 and 3 differ). For a prime power $q \geq 3$ both theorems give the same bound $N(4q, 5) \geq 144\, q^5$. As a lower bound, Theorem 3 adds nothing to Corollary 2. The two theorems differ in the graph. Theorem 1 uses *some* commutative ring with q elements. In our proof this is a product of copies of $\mathbb{F}_4$, $\mathbb{F}_8$ and $\mathbb{Z}/p\mathbb{Z}$, for example $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ for $q = 9$ and $\mathbb{F}_8 \times \mathbb{F}_4$ for $q = 32$. Theorem 3 uses the field $\mathbb{F}_q$ itself, that is, the construction of arXiv:2606.15860 with $\mathbb{F}_q$ in place of $\mathbb{F}_3$ or $\mathbb{F}_4$. For q prime and for $q = 4, 8$ the two graphs coincide; they differ for $q = 9, 16, 25, 27, 32, \ldots$

The proof of Theorem 3 needs three further ingredients (Section 6): field extensions; explicit charts over $\mathbb{F}_{32}$, $\mathbb{F}_{128}$, $\mathbb{F}_{2048}$ and $\mathbb{F}_{8192}$; and a Schwartz–Zippel argument for the remaining fields $\mathbb{F}_{2^m}$. Theorem 1 avoids all three by passing to products of rings, so that only $\mathbb{F}_4$, $\mathbb{F}_8$ and $\mathbb{Z}/p\mathbb{Z}$ are needed. This is why Theorem 1, and not Theorem 3, is formalized in Lean.

Proposition 5. *There is no universal route chart on $\mathbb{F}_2^5$. Consequently no finite commutative ring of order $n \equiv 2 \pmod 4$ carries a universal route chart, and Theorem 1 is the full range of orders reached by route-chart lifts over finite commutative rings.*

Theorem 1 is proved in Sections 2–5 and is formalized in Lean (Section 8). Theorem 3 and Proposition 5 are proved in Sections 6 and 7. They rely on further computer checks, not on the Lean formalization.

2 Route charts over commutative rings

Throughout, R is a finite commutative ring. Let $S = \{a, a^{-1}, b, b^{-1}\}$, with the involution $t \mapsto t^{-1}$. A word $t_1t_2t_3t_4t_5$ over S is *reduced* if $t_{i+1} \neq t_i^{-1}$ for $1 \leq i \leq 4$; there are $4 \cdot 3^4 = 324$ reduced words of length 5.

Definition 6 (Controller)**.** A *controller* on a finite set C is a family of maps $\sigma_t \colon C \to C$ $(t \in S)$ with $\sigma_{t^{-1}}(\sigma_t(u)) = u$ for all t and u. It is *exact-NB*-5 if for all $u, v \in C$ there is a reduced word $t_1 \cdots t_5$ with $\sigma_{t_5} \cdots \sigma_{t_1}(u) = v$.

The controller of arXiv:2606.15860, §A.5, is a simple 4-regular graph on 144 vertices. Its arcs are labelled by S so that every vertex has exactly one outgoing arc with each label and the reverse of an arc with label t has label t^{-1} (§A.3). Writing $\sigma_t(u)$ for the head of the arc with label t leaving u gives a controller in the above sense. A walk is nonbacktracking exactly when its label word is reduced. The paper verifies that this controller is exact-NB-5, and we verify it again (Section 8).

Definition 7 (Route chart)**.** A *route chart* on R^5 assigns to every $t \in S$ a matrix $A_t \in M_5(R)$ and a vector $b_t \in R^5$ with

$$A_{t^{-1}} A_t = I, \qquad b_{t^{-1}} = A_{t^{-1}}\, b_t \qquad (t \in S). \tag{1}$$

For a word $w = t_1 \cdots t_5$ let

$$M_w = \big[\, A_{t_5} A_{t_4} A_{t_3} A_{t_2} b_{t_1} \;\big|\; A_{t_5} A_{t_4} A_{t_3} b_{t_2} \;\big|\; A_{t_5} A_{t_4} b_{t_3} \;\big|\; A_{t_5} b_{t_4} \;\big|\; b_{t_5} \,\big] \in M_5(R). \tag{2}$$

The chart is *universal* if M_w has a right inverse for every reduced word w of length 5. Over a commutative ring this means that $\det M_w$ is a unit, and over a field that M_w is nonsingular, as in arXiv:2606.15860.

Definition 8 (Lift graph)**.** Given a controller on C and a route chart on R^5, the graph G has vertex set $C \times R^5$. Two distinct vertices (u, x) and (v, y) are adjacent if $v = \sigma_t(u)$ and $y = A_t x + \lambda b_t$ for some $t \in S$ and $\lambda \in R$.

This relation is symmetric. If $v = \sigma_t(u)$ and $y = A_t x + \lambda b_t$, then $\sigma_{t^{-1}}(v) = u$ and, by (1), $A_{t^{-1}} y + (-\lambda) b_{t^{-1}} = x + \lambda A_{t^{-1}} b_t - \lambda b_{t^{-1}} = x$. When the controller is a simple graph and R is a field, this is the route-chart lift graph of arXiv:2606.15860, §A.4.

Lemma 9. *(a) Every vertex of G has at most $4|R|$ neighbours.*

(b) If the controller is exact-NB-5 and the chart is universal, every two vertices of G are joined by a walk of length at most 5.

Proof. (a) The neighbours of (u, x) lie in $\{(\sigma_t(u), A_t x + \lambda b_t) : t \in S,\ \lambda \in R\}$.

(b) Let (u, x) and (v, y) be vertices. Choose a reduced word $w = t_1 \cdots t_5$ with $\sigma_{t_5} \cdots \sigma_{t_1}(u) = v$, and a right inverse N of M_w. Put $A_w = A_{t_5} \cdots A_{t_1}$ and $\lambda = N(y - A_w x) \in R^5$. Define $x_0 = x$, $u_0 = u$ and, for $i = 1, \ldots, 5$, $u_i = \sigma_{t_i}(u_{i-1})$ and $x_i = A_{t_i} x_{i-1} + \lambda_i b_{t_i}$. Expanding,

$$x_5 = A_w x + \sum_{j=1}^{5} \lambda_j \, A_{t_5} \cdots A_{t_{j+1}} b_{t_j} = A_w x + M_w \lambda = A_w x + (y - A_w x) = y,$$

and $u_5 = v$. Consecutive vertices (u_{i-1}, x_{i-1}) and (u_i, x_i) are equal or adjacent, so deleting repetitions from $(u_0, x_0), \ldots, (u_5, x_5)$ gives a walk of length at most 5 from (u, x) to (v, y). □

3 Transport, products and the orders reached

Lemma 10. *Let $f\colon R \to R'$ be a homomorphism of commutative rings. Applying f to all entries maps a route chart on R^5 to a route chart on R'^5, maps each M_w to the corresponding matrix of the image chart, and preserves universality.*

Proof. Entrywise application of f commutes with sums and products of matrices and vectors, so (1), (2) and the equations $M_w N = I$ are preserved. □

Lemma 11. *If R_1 and R_2 carry universal route charts, so does $R_1 \times R_2$.*

Proof. Pair the two charts entrywise: $A_t = (A_t^{(1)}, A_t^{(2)})$ and $b_t = (b_t^{(1)}, b_t^{(2)})$. All identities hold componentwise, and the pair $(N^{(1)}, N^{(2)})$ of right inverses is a right inverse of M_w. □

Call n *good* if some commutative ring with n elements carries a universal route chart. The zero ring shows that 1 is good, since every matrix over it is the identity. By Lemma 11, if m and n are good then so is mn.

Lemma 12. *If 4, 8 and every odd prime are good, then every $n \geq 1$ with $n \not\equiv 2 \pmod 4$ is good.*

Proof. Strong induction on n. The case $n = 1$ is done. If $n > 1$ is odd, write $n = pk$ with p the smallest prime factor of n. Then p is odd and $k < n$ is odd, so n is good. If n is even, then $4 \mid n$. Write $n = 4k$. If $k \not\equiv 2 \pmod 4$, use k. Otherwise $n = 8m$ with $m = k/2$ odd, and we use m. □

4 Charts for the building blocks

All charts below are given in the data files of `general-q/data/charts` (Section 8).

$\mathbb{F}_4$ and $\mathbb{F}_8$. The chart on $\mathbb{F}_4^5$ is the chart of arXiv:2606.15860, §A.6.2. The chart on $\mathbb{F}_8^5$ was found by simulated annealing. For each of the 324 reduced words, an inverse of M_w is recorded, and the products $M_w N_w = I$ are checked.

$\mathbb{Z}/p\mathbb{Z}$ **for odd primes** $p \le 79$**.** For $p = 3$ we use the chart of arXiv:2606.15860, §A.6.1. For $p = 43$ we use the reduction modulo 43 of integer chart 1 below. Every other odd prime $p \le 79$ has a chart found by simulated annealing. Each comes with inverse certificates as above.

Primes $p > 79$**.** We use six *integer charts*: matrices $A_a, A_b \in \mathrm{GL}_5(\mathbb{Z})$ of determinant ± 1 with integer inverses, integer vectors b_a, b_b, and $A_{t^{-1}} = A_t^{-1}$, $b_{t^{-1}} = A_{t^{-1}} b_t$, so that (1) holds over $\mathbb{Z}$. For every reduced word w, an integer d_w and an integer matrix adj_w are recorded with $M_w \,\mathrm{adj}_w = d_w I$; here $d_w = \det M_w \neq 0$ and adj_w is the adjugate. If a prime p divides none of the d_w of one chart, the reduction of that chart modulo p is universal by Lemma 10: d_w is a unit of $\mathbb{Z}/p\mathbb{Z}$, and $d_w^{-1}\mathrm{adj}_w$ is a right inverse of M_w. Let $P_i = \prod_w |d_w|$ for chart i and

$$g = \gcd(P_1, \ldots, P_6) = 2^{338}\, 3^{146}\, 5^{74}\, 7^{34}\, 11^{24}\, 13^{22}\, 17^{10}\, 19^{12}\, 23^{10} \cdot 29^2\, 31^2\, 37^4\, 41^2\, 47^2\, 53^2\, 59^4\, 61^2\, 67^2\, 71^2\, 73^2\, 79^4.$$

A prime $p > 79$ does not divide g, so it does not divide some P_i, and chart i works modulo p. (The prime 43 does not divide g either.)

Corollary 13. *4, 8 and every odd prime are good.*

5 Proof of Theorem 1

By Corollary 13 and Lemma 12, every $n \ge 1$ with $n \not\equiv 2 \pmod 4$ is good. Let R be a ring with n elements carrying a universal chart, and let C be the 144-vertex controller, which is exact-NB-5. By Lemma 9, the lift graph on $C \times R^5$ has $144\, n^5$ vertices and maximum degree at most $4n$, and every two vertices are joined by a walk of length at most 5. □

Explicitly, R is a product of copies of $\mathbb{F}_4$ and $\mathbb{F}_8$ (at most one $\mathbb{F}_8$) and of $\mathbb{Z}/p\mathbb{Z}$ for odd primes p dividing n. For $n = q$ a prime power one may instead take the field $\mathbb{F}_q$; this is Theorem 3.

6 The field version: proof of Theorem 3

By Lemma 10 applied to the inclusion of a subfield, a universal chart on K^5 is universal on L^5 for every field extension $L \supseteq K$.

- $q = p^m$ with p odd: $\mathbb{F}_q \supseteq \mathbb{F}_p$, and $\mathbb{F}_p$ carries a universal chart by Section 4.
- $q = 2^m$ with m even: $\mathbb{F}_q \supseteq \mathbb{F}_4$.
- $q = 2^m$ with m odd and a prime factor $r \in \{3, 5, 7, 11, 13\}$: $\mathbb{F}_q \supseteq \mathbb{F}_{2^r}$. Universal charts on $\mathbb{F}_8^5$, $\mathbb{F}_{32}^5$ and $\mathbb{F}_{128}^5$ were found by simulated annealing, and on $\mathbb{F}_{2048}^5$ and $\mathbb{F}_{8192}^5$ by random sampling. Each is checked on all 324 reduced words.
- $q = 2^m$ with every prime factor of m at least 17: then $q \ge 2^{17}$, and we use the Schwartz–Zippel lemma as follows.

Treat the 60 entries of A_a, A_b, b_a, b_b as indeterminates and set $A_{t^{-1}} = \mathrm{adj}(A_t)/\det A_t$ and $b_{t^{-1}} = \mathrm{adj}(A_t) b_t / \det A_t$ for $t \in \{a, b\}$. Multiply column j of M_w by $\det A_a$ once for each occurrence of a^{-1} among $t_{j+1}, \ldots, t_5$, and once more if $t_j = a^{-1}$; do the same for b. The result has polynomial entries. Each factor A_t contributes degree 1 and each $\mathrm{adj}(A_t)$ degree 4, and the last factor b_t or $\mathrm{adj}(A_t) b_t$ has degree at most 5. So column j has degree at most $4(5 - j) + 5$. The determinant D_w

of the modified matrix therefore has degree at most $\sum_{j=1}^{5}\big(4(5-j)+5\big) = 65$, and it equals $\det M_w$ times powers of $\det A_a$ and $\det A_b$. The polynomial

$$P = \det A_a \cdot \det A_b \cdot \prod_w D_w$$

has degree at most $10 + 324 \cdot 65 = 21{,}070$ and integer coefficients. It does not vanish at the $\mathbb{F}_4$ chart, so its reduction modulo 2 is a nonzero polynomial over $\mathbb{F}_2$. For $q = 2^m > 21{,}070$, the Schwartz–Zippel lemma gives a point of $\mathbb{F}_q^{60}$ where $P \neq 0$. There A_a and A_b are invertible and every $\det M_w \neq 0$, which is a universal chart on $\mathbb{F}_q^5$.

The graph statement follows from Lemma 9 and the Moore bound remark after Corollary 2. The graph is $4q$-regular: the controller is simple, so distinct labels lead to distinct controller vertices, and $b_t \neq 0$ because b_t is the last column of the nonsingular matrix M_{ttttt}. □

7 The case $\mathbb{F}_2$: proof of Proposition 5

Suppose (A_t, b_t) is a universal chart on $\mathbb{F}_2^5$. Since M_{aaaaa} is nonsingular, b_a is a cyclic vector of A_a, so A_a is cyclic. Simultaneous conjugation by $Q \in \mathrm{GL}_5(\mathbb{F}_2)$, that is $(A_t, b_t) \mapsto (QA_tQ^{-1}, Qb_t)$, preserves universality because $M_w \mapsto QM_w$. So we may assume that A_a is the companion matrix of its characteristic polynomial f, of degree 5 with $f(0) = 1$ (16 possibilities). The centralizer of a cyclic matrix acts simply transitively on its cyclic vectors, so we may further assume that $b_a = e_0$. An exhaustive search over all 9,999,360 matrices $A_b \in \mathrm{GL}_5(\mathbb{F}_2)$ and all 31 nonzero b_b, for each f, finds no universal chart (`scripts/f2exhaust.c`). Its evaluation of the 324 words was cross-checked against an independent implementation on random charts.

A finite commutative ring is a product of finite local rings, each of prime power order. If $|R| \equiv 2 \pmod 4$, exactly one factor has even order, and that order is 2, so the factor is $\mathbb{F}_2$. The projection $R \to \mathbb{F}_2$ would map a universal chart on R^5 to one on $\mathbb{F}_2^5$ (Lemma 10). □

8 Data, verification and the Lean formalization

The folder `general-q`, inside `results/diameter5-paper` of the repository `https://github.com/rmizuno42/ddp` next to the materials of arXiv:2606.15860, contains:

- `data/controller.json`: the arc maps σ_t of the 144-vertex controller, exported from the paper's supplementary code, and for every ordered pair of vertices a reduced word of length 5 joining them.
- `data/charts/`: the charts over $\mathbb{F}_4$, $\mathbb{F}_8$, $\mathbb{Z}/p\mathbb{Z}$ ($3 \le p \le 79$), the six integer charts, and the extension-field charts of Section 6.
- `scripts/`: independent Python checks of the charts, the certificate generator, the search programs, and the $\mathbb{F}_2$ search.
- `lean/`: the Lean 4 formalization of Theorem 1.

Lean. The formalization (Lean 4.32.2, Mathlib) defines controllers, route charts over arbitrary commutative rings and lift graphs, and proves Lemmas 9–12 and the argument for primes $p > 79$ in general. The finite facts are proved by `decide +kernel`, that is, evaluated by the Lean kernel itself:

- the controller identities $\sigma_{t^{-1}}\sigma_t = \mathrm{id}$, and the certificate words for all 144^2 ordered pairs;
- for each chart, relations (1) and, for every reduced word, the certificate identity $M_w N_w = I$ (or $M_w \operatorname{adj}_w = d_w I$);
- the factorization of g.

To keep the kernel computation small, the checks run on lists of natural numbers or integers, and lemmas proved once show that this list arithmetic agrees with the corresponding matrix operations over the ring. $\mathbb{F}_4$ and $\mathbb{F}_8$ are defined by their addition and multiplication tables, and their ring axioms are checked by `decide`. The main theorem is `DDP.main` (and `DDP.main_prime_power` for prime powers). It has no hypotheses beyond its statement, and `native_decide` is not used. A GitHub Actions workflow builds the whole project and checks with `#print axioms` that both theorems depend only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`. Theorem 3 and Proposition 5 are not formalized.

The case $n = 5$. The lift over $\mathbb{Z}/5\mathbb{Z}$ (equal to $\mathbb{F}_5$) has 450,000 vertices and 4,500,000 edges. Its edge list was checked with the repository's exact bit-parallel breadth-first search from every vertex: it is 20-regular, connected, and of diameter 5. The same construction program reproduces the published $(12, 5)$ and $(16, 5)$ edge lists of arXiv:2606.15860 exactly.

Comparison. The table of F. Comellas covers $\Delta \leq 16$. In the CombinatoricsWiki degree–diameter table (last modified 18 February 2022) the entry for $(20, 5)$ is 281,820. We have not compared with later results. For $\Delta = 4n \geq 28$ we know of no table, and we have not reviewed the literature on large-degree constructions of diameter 5; the family gives $N(\Delta, 5) \geq (144/1024)\,\Delta^5 \approx 0.14\,\Delta^5$ for these Δ.

Acknowledgements

The controller and the charts over $\mathbb{F}_3$ and $\mathbb{F}_4$ come from arXiv:2606.15860, where they were found through interaction with ChatGPT (OpenAI). The extension in this note, the chart searches, the certificates and the Lean formalization were produced with Claude Code (Anthropic) under the author's direction.

References

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